

Direct-to-Phase-II (DP2) Proposal

Topic A254-023: AI-Enabled Portfolio Management

1. Executive Summary

[Company Name] proposes **AcquiSense-AI**, an AI-powered Acquisition Requirements Intake and Portfolio Management platform tailored to the U.S. Army's unique budget, appropriations, and regulatory environment. Leveraging advanced machine learning, natural language processing (NLP), and business-process optimization (Lean Six Sigma), AcquiSense-AI will:

- **Automate & standardize** intake of new IT requirements via configurable templates and guided workflows.
- **Intelligently prioritize** and sequence requirements against qualification criteria, existing capabilities, and contract inventories.
- **Provide real-time visibility** into requirement status and portfolio health through an integrated dashboard.
- **Continuously scan** internal and external contract repositories to detect redundancy and identify opportunities for reuse.

Having completed an SBIR-equivalent Phase I in partnership with a Tier 1 DoD lab—delivering a functional prototype that ingests requirement requests, applies basic NLP classification, and correlates them against a mock contract database—our team is ready to transition directly into Phase II. Within 120 days of award, we will deliver a first MVP; over the full 18-month period, we will expand to a fully featured, production-ready platform.

2. Phase I Equivalency & Supporting Evidence

To satisfy DP2 prerequisites, we submit:

1. **Prototype Demonstration Videos & Code:** A 10-minute walkthrough showing our current proof-of-concept portal handling intake forms, executing NLP-based tagging, and matching against a sample portfolio of 500 contracts.
2. **Performance Metrics:**
 - 92 % accuracy in automated requirement-to-contract matching (measured on a 200-item test set).

- Intake processing latency under 2 seconds per submission.
 - 3. **Technical Reports:** Documentation of our Lean Six Sigma-driven process redesign, system architecture diagrams, and detailed test logs.
 - 4. **Customer Letter of Support:** From a DoD acquisition office validating the prototype's relevance and performance.
-

3. Technical Objectives (Phase II)

1. MVP 1 (Day 120)

- Core intake portal with configurable workflows and digital forms.
- NLP engine for automated classification (e.g., capability area, acquisition category).
- Contract/asset database connector with basic search and match.
- User dashboard showing intake queue and match suggestions.

2. MVP 2 (Month 6)

- Advanced AI prioritization using multi-criteria decision analysis (MCDA).
- Automated governance rules engine for approve/route/escalate.
- Integration with Army's existing systems (e.g., CMP, procurement DB).

3. MVP 3 (Month 12)

- Continuous external scanning module (FedBizOps, GSA schedules).
- Portfolio analytics: redundancy detection, cost-avoidance reporting.
- Collaboration features: comment threads, notifications, audit logs.

4. Final Release (Month 18)

- Full feature set hardened for Authority to Operate (ATO) in AWS GovCloud.
 - Comprehensive training materials, admin console, and sustainment plan.
-

4. Technical Approach

1. Lean Six Sigma Process Redesign

- Map “as-is” intake workflows, identify bottlenecks, and design “to-be” processes.
- Facilitate stakeholder workshops to codify templates and governance criteria.

2. AI/ML & NLP Pipeline

- Fine-tune transformer-based models (e.g., RoBERTa) on labeled Army requirement data.
- Develop MCDA models combining technical merit, cost, schedule, and strategic fit.
- Implement feedback loops for continuous model retraining using user corrections.

3. System Architecture & Cloud Infrastructure

- Containerized microservices on AWS GovCloud (EKS, RDS, Elasticsearch).
- Secure API layer for integration with external systems (OAuth 2.0, TLS 1.2+).
- Role-based access control (RBAC) and audit logging for compliance.

4. User Experience & Change Management

- Employ human-centered design to ensure usability across business and technical roles.
- Develop communications plan and training modules aligned with organizational change management best practices.

5. Work Plan & Milestones

Timeframe	Key Activities	Milestone Deliverable
Months 0–1	Project kickoff; detailed requirements validation; data ingestion	Detailed Project Plan & Data Schema
Months 2–4	Develop intake portal, initial NLP models, contract DB connector	MVP 1 Demonstration (Day 120)
Months 5–7	Enhance AI prioritization, governance engine,	MVP 2 Release

Timeframe	Key Activities	Milestone Deliverable
	integrate DoD systems	
Months 8–11	Build external scanning, portfolio analytics, collaboration features	MVP 3 Release
Months 12–17	Security hardening, user training, ATO prep, performance tuning	Pre-release Candidate
Month 18	Final testing, documentation, transition to sustainment	Final DP2 Deliverables & ATO Approval

6. Key Personnel

Name	Role	Relevant Expertise
Dr. Morgan Lee	Principal Investigator	15 yrs AI/ML in defense acquisition; ex-DARPA program manager
Ms. Elena Ruiz	Acquisition SME	12 yrs Army acquisition advisor; Lean Six Sigma Black Belt
Mr. Sanjay Mehta	Lead Data Scientist	10 yrs NLP & ML model development; transformer architectures
Ms. Allison Parker	Systems Architect	8 yrs AWS GovCloud, microservices, security compliance
Mr. Thomas Zhang	Change Management Lead	7 yrs FedGov organizational change; training & communications specialist

7. Company Overview

[Company Name] is a certified small business with CMMC Level 3, FedRAMP Ready status, and ITAR-compliant facilities. We specialize in AI-driven defense acquisition solutions and have successfully delivered multiple SBIR and commercial projects in AWS GovCloud. Our dedicated DevSecOps pipeline and TS/SCI-cleared personnel ensure secure, rapid delivery.

8. Facilities & Environment

- **Secure Development Lab:** Segmented networks, hardware firewalls, vulnerability scanners.
 - **Compute Resources:** GPU clusters for model training, EKS clusters for deployment.
 - **Collaboration Tools:** Jira/Confluence, Microsoft Teams with DoD accreditation.
-

9. DP2 Deliverables

- **Day 120:** MVP 1 portal, NLP classification, contract matching, user dashboard.
 - **Month 6:** MVP 2 advanced prioritization, governance engine, DoD integrations.
 - **Month 12:** MVP 3 analytics, external scanning, collaboration suite.
 - **Month 18:** Production-ready platform with ATO, training, sustainment plan.
-

10. Phase III & Dual-Use Opportunities

Post-DP2, AcquiSense-AI will be positioned for enterprise-wide Army rollout (TRL 9) and commercial adaptation for highly regulated sectors (finance, healthcare), leveraging its modular AI framework and robust compliance foundation.